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18 April 1979
SUPERSEDING
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15 March 1977

MILITARY SPECIFICATION

SWITCH, BALANCED MAGNETIC, SA-1955/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a balanced magnetic switch used to detect the unauthorized opening of a door, window, or other entrance into an area which has been secured.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-22751	- Coating System, Epoxy-Polyamide, Chemical and Solvent Resistant: Process for Application of.
MIL-C-45662	- Calibration System Requirements.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E3300	- Switch, Balanced Magnetic, SA-1955/FSS-9(V).
TA13220E3301	- Magnet Assembly, Actuating.
TA13220E3304	- Printed Wiring Board Assembly.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. The balanced magnetic switch (hereinafter referred to as BMS) shall be in accordance with Top Assembly TA13220E3300 and as specified herein.

3.1.1 Major components. The major components of the BMS are as follows:

(a) Magnetic Assembly, Actuating, TA13220E3301.
(b) Switch Assembly, Magnetic.
    (1) Printed Wiring Board (PWB) Assembly, TA13220E3304.
    (2) Bias Magnet Assembly.
    (3) Switch Assembly Enclosure.

When specified (see 6.2), any of the above major components shall be furnished separately.

3.1.2 Function. The BMS detects an intrusion whenever a window or door or other type of access on which the BMS is installed is opened. The actuating magnet is installed on the movable surface. The Magnetic Switch Assembly is installed on the non-moving surface and adjusted for magnetic balance. When the actuating magnet is moved or disturbed, the switch is activated to the alarm state.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more BMS's and major components (when procured separately) as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce BMS's and major components (when procured separately) that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests

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shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more BMS's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.5 Environmental conditions. Each assembly shall withstand the individual or a combination of the environmental conditions specified herein without mechanical or electrical damage (see 6.4.2), or performance degradation.

3.5.1 Operational.

(a) Temperature range of minus 65° F to plus 160° F.
(b) Impact testing as specified in 4.6.2.6.6.
(c) Adverse weather - rain or salt fog when tested as specified in 4.6.2.6.7 and 4.6.2.6.8.

3.5.2 Nonoperational and storage.

(a) Temperature range of minus 80° F to plus 165° F.
(b) Humidity up to 95 percent relative.
(c) Shock of 20 times the acceleration due to gravity (g's) for 11 milliseconds (ms) and bench handling.
(d) Vibration - transportation handling.

3.6 Detailed technical requirements.

3.6.1 General. The BMS shall transmit an alarm state to the interconnecting wiring whenever the Actuating Magnet Assembly is moved beyond a specified distance relative to the Magnetic Switch Assembly, or under any conditions which result in a specified magnetic field change at the two assemblies. A tamper alarm shall be transmitted whenever the cover to the Magnetic Switch Assembly is opened.

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3.6.2 Terminal board connections. The terminal board (TB) shall be marked with identifying terminal numbers as shown:

TB1	Function
1	Alarm switch "N.O." contact (with actuating magnet removed from vicinity of alarm switch).
2	Alarm switch common contact.
3	Spare.
4	Spare.
5	Tamper switch contact.
6	Tamper switch contact.
7	Alarm switch "N.C." contact (with actuating magnet removed from vicinity of alarm switch).

A removable jumper shall be supplied across Terminals 2 and 3, 3 and 4, and 4 and 5.

3.6.3 Functional characteristics.

3.6.3.1 Alarm condition. The alarm condition shall be represented by an internal impedance of more than 100 kilohms (K ohms) across Terminals 1 and 2, and less than 37 ohms across Terminals 2 and 7 of TB1. The nonalarm condition shall be represented by an internal impedance of 33 $\pm$ 4 ohms across Terminals 1 and 2 and more than 100K ohms across Terminals 2 and 7 of TB1. The major components shall function when the Switch Assembly Enclosure and the Actuating Magnet Assembly are installed on a common plane with separations of 0.250 inch to 1 inch. For any spacing in this range, an alarm condition shall not be transmitted when the actuating magnet is moved away from the Magnetic Switch Assembly less than 0.250 inch (movement shall be normal to the mounting plane). The alarm condition shall be transmitted when the actuating magnet is moved farther away from the switch assembly. This condition shall occur before a distance of 1.250 inches is reached. An adjustable biasing magnet shall be employed to achieve these requirements. When installed in the no-alarm state, an alarm condition shall be transmitted when an external magnetic field is induced at the Magnetic Switch Assembly enclosure. The total field variation, determined by the addition of the absolute values of the positive and negative polarity field strengths, shall not exceed 67 gauss.

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3.6.3.2 Tamper protection. A tamper condition shall be represented by an impedance exceeding 100K ohms between Terminals 5 and 6 of TB1. The nontamper condition shall be represented by an impedance of less than 1 ohm between these terminals. The tamper condition shall be transmitted whenever the cover of the Magnetic Switch Assembly is opened from 0.087 inch to 0.262 inch.

3.6.4 Life. When tested as specified in 4.6.2.7, the BMS shall be capable of not less than 1,000,000 operations.

3.6.5 Printed wiring board assembly. The PWB Assembly shall be in accordance with applicable Top Assembly TA13220E3304 and have the following characteristics:

Terminals	Action	Resistance
TB1-1 to TB1-2	None	33 ± 4 ohms
TB1-2 to TB1-7	None	>100K ohms
TB1-2 to TB1-3	None	<1 ohm
TB1-2 to TB1-4	None	<1 ohm
TB1-2 to TB1-5	None	<1 ohm
TB1-2 to TB1-6	None	>100K ohms
TB1-2 to TB1-6	Depress S1	<1 ohm
TB1-1 to TB1-2	S2 Energized	>100K ohms
TB1-2 to TB1-7	S2 Energized	33 ± 4 ohms

When the PWB Assembly is installed in the Switch Assembly Enclosure with the Bias Magnet Assembly (see Drawing C13220E3305) installed on the PWB, the PWB shall meet the requirements of 3.6.3.1 and 3.6.3.2.

3.6.6 Actuating Magnet Assembly. The Actuating Magnet Assembly shall be in accordance with applicable Top Assembly TA13220E3301. The magnetic induction at the end of the magnet in housing (see Drawing C13220E3311-1) shall be 1000 to 1700 gauss and in housing (see Drawing C13220E3311-2) shall be 1500 to 2000 gauss.

3.7 Identification marking. The BMS and parts shall be identified in accordance with MIL-STD-130.

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3.8 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the BMS and major components normally painted shall be cleaned, treated, and painted in accordance with MIL-C-22751. Color shall be lusterless gray conforming to FED. STD. No. 595, Color No. 36440. Nonpermissible galvanic couples (dissimilar metals) as specified in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.9 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5) for testing major components when procured separately.

Item No.	Description	Identification	Quantity
1.	Switch, Balanced Magnetic, SA-1955/FSS-9(V)	TA13220E3300	1

3.10 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.11 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. These requirements are applicable to wiring, welding and brazing, plating, riveting, finishes, machine operation, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part or equipment unsatisfactory for the purpose intended.

3.11.1 Soldering. All soldering shall be in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requrements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with the specifications, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.3 and 6.8).

4.2 Classification of inspections. The inspection requirements specified herein are classified below. Individual testing of major components installed in the BMS is not required. Major components procured as separate items shall be individually inspected as specified herein.

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

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4.3 Preproduction inspection.

4.3.1 Examination. The preproduction BMS and major components (when procured separately) shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 Complete BMS. The complete BMS shall be subjected to the tests marked "X" in Column 1 of Table I. Individual testing of major components of the complete BMS is not required. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Major components procured separately. Major components (when procured separately) shall be subjected to the tests marked "X" in Column 1 of Tables II and III. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production BMS's will be selected at random by the Government from BMS's produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production BMS will not exclude the remaining BMS's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test. Failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production BMS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production BMS and shall be cause for refusal by the Government to continue acceptance of production BMS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be

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accomplished by the contractor at no cost to the Government on BMS's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all BMS's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Each BMS and major component (when procured separately) shall be examined for defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual.

4.5.3.1.1 Complete BMS. Each complete BMS shall be subjected to the tests marked "X" in Column 3 of Table I. Individual testing of major components of the complete BMS is not required. Failure of any test shall be cause for rejection.

4.5.3.1.2 Major components procured separately. Each major component (when procured separately) shall be subjected to the tests marked "X" in Column 2 of Tables II and III. Failure of any test shall be cause for rejection.

4.5.3.2 Samples.

4.5.3.2.1 Complete BMS. Complete BMS shall be subjected to the tests marked "X" in Column 4 of Table I. Individual testing of major components of the complete BMS is not required. AQL shall be 2.5 percent defective.

4.5.3.2.2 Major components procured separately. Major components (when procured separately) shall be subjected to the tests marked "X" in Column 3 of Tables II and III. AQL shall be 2.5 percent defective.

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4.6 Inspection procedure.

4.6.1 Examination. The BMS and major components (when procured separately) shall be examined as specified for the following defects:

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.
104. Identification or special marking incorrect, missing, or incomplete.
105. Treatment and painting not as specified.
106. Interchangeability not as specified.
107. Workmanship, including wiring and soldering, not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, the tests shall be performed under the ambient climatic conditions existing at the place of the test. Unless otherwise specified herein, the BMS and major components (when procured separately) shall operate without maintenance. Test equipment shall be calibrated in accordance with MIL-C-45662.

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4.6.2.2 Test schedule.

TABLE I. Test schedule for complete BMS.

Preproduction	Initial production	Quality conformance: Individual	Quality conformance: Sample	Test	Requirement paragraph	Test paragraph
1	2	3	4	5	6	7
				Functional:		
X	X	X	-	(a) Alarm condition	3.6.3.1	4.6.2.3.1
X	X	X	-	(b) Tamper protection	3.6.3.2	4.6.2.3.2
X	X	-	X	Low temperature	3.5.1, 3.5.2	4.6.2.6.1
X	X	-	X	High temperature	3.5.1, 3.5.2	4.6.2.6.2
X	X	-	-	Humidity	3.5.2	4.6.2.6.3
X	X	-	-	Shock	3.5.2	4.6.2.6.4
X	X	-	-	Vibration	3.5.2	4.6.2.6.5
X	X	-	-	Impact	3.5.1	4.6.2.6.6
X	X	-	-	Rain	3.5.1	4.6.2.6.7
X	X	-	-	Salt fog	3.5.1	4.6.2.6.8
X	X	-	-	Life	3.6.4	4.6.2.7

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TABLE II. Test schedule for PWB Assembly.[1/]

Preproduction	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional[2/]	3.6.5	4.6.2.4
X	-	X	Low temperature[2/]	3.5.1, 3.5.2	4.6.2.6.1
X	-	X	High temperature[2/]	3.5.1, 3.5.2	4.6.2.6.2
X	-	X	Humidity[2/]	3.5.2	4.6.2.6.3
X	-	-	Shock[2/]	3.5.2	4.6.2.6.4
X	-	-	Vibration[2/]	3.5.2	4.6.2.6.5

1/ The PWB Assembly shall be tested with the Bias Magnet Assembly emplaced on the PWB forming an integral unit.
2/ Place PWB Assembly on Government-loaned BMS or equivalent to simulate normal environment when installed in the BMS.

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TABLE III. Test schedule for Actuating Magnet Assembly.

Preproduction	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.6	4.6.2.5
X	-	X	Low temperature	3.5.1, 3.5.2	4.6.2.6.1
X	-	X	High temperature	3.5.1, 3.5.2	4.6.2.6.2
X	-	X	Humidity	3.5.2	4.6.2.6.3
X	-	-	Shock	3.5.2	4.6.2.6.4
X	-	-	Vibration	3.5.2	4.6.2.6.5

4.6.2.3 Functional.

4.6.2.3.1 Alarm condition. The BMS shall be tested in accordance with Appendix Test Method 1, Procedure (a). Nonconformance to 3.6.3.1 shall constitute failure of this test.

4.6.2.3.2 Tamper protection. The BMS shall be tested in accordance with Appendix Test Method 1, Procedure (b). Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.6.2.4 Printed Wiring Board Assembly. The PWB Assembly shall be tested in accordance with Appendix Test Method 1. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.2.5 Actuating Magnet Assembly. The Actuating Magnet Assembly shall be tested in accordance with Appendix Test Method 1. Nonconformance to 3.6.6 shall constitute failure of this test.

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4.6.2.6 Environmental.

4.6.2.6.1 Low temperature. Low temperature test as specified in MIL-STD-810, Method 502.1, Procedure I, storage temperature minus 80° F for not less than 48 hours, equipment operating temperature minus 65° F.

4.6.2.6.2 High temperature. High temperature test as specified in MIL-STD-810, Method 501.1, Procedure II, except that the nonoperating temperature shall be plus 165° F. The operating temperature shall be plus 160° F.

4.6.2.6.3 Humidity. Humidity test as specified in MIL-STD-810, Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Perform measurements at the completion of each cycle.
(b) During Steps 2 and 3, relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. For the Tamper Test the cover of the BMS may be wiped of excess external moisture prior to the performance of this test. This test may be the last functional test performed prior to continuation of humidity cycling. The BMS may be removed from the environmental chamber for measurement; however unsuccessful completion of measurements or unsuccessful return of the BMS to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of this test.

4.6.2.6.4 Shock.

(a) Basic design: As specified in MIL-STD-810, Method 516.2, Procedure I, equipment not operating, sawtooth shock pulse, peak value 20 g's, duration 11 ms.
(b) Bench handling: As specified in MIL-STD-810, Method 516.2, Procedure V.

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4.6.2.6.5 Vibration. Vibration test as specified in MIL-STD-810, Method 514.2, Procedure X, Curve AX, equipment not operating.

4.6.2.6.6 Impact. Subject the BMS to the tests specified in MIL-S-901, Grade A, hull mounted, Equipment Class I, Test Class: Lightweight, Type A, deenergized.

4.6.2.6.7 Rain. Rain test as specified in MIL-STD-810, Method 506.1, Procedure I.

4.6.2.6.8 Salt fog. Salt fog test as specified in MIL-STD-810, Method 509.1, operating, test at each 24 hour interval.

4.6.2.6.9 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

4.6.2.7 Life test. The BMS's shall be tested initially as specified in 4.6.2.3. Then, multiple units shall be subjected to a life test for a total of 1,000,000 operation cycles under a load of 250 milliamperes (mA) driven from a 48 volts direct current (Vdc) source (applied between Pins 1 and 2 of TB1). The operation cycle shall be at a rate of approximately one per second. The test shall be conducted under ambient uncontrolled laboratory conditions, without temperature, humidity, or vibration cycling. Operation shall be continuous throughout the test. During each 8 hour period of testing, subject the BMS's to the tests specified in 4.6.2.3. Failure of the BMS's to function or nonconformance to 3.6.3.1 or 3.6.3.2 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select BMS's at any time during the contract production period and subject these BMS's to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on BMS's selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production BMS's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison BMS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison BMS's and shall be cause for refusal by the Government to continue acceptance of production BMS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on BMS's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all BMS's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
109.	Components not preserved by the applicable procedure for Level A or Level B.
110.	Boxed contents not cushioned to prevent movement or damage as specified for Level A or Level B.
111.	Consolidation not as specified for Level A or Level B.
112.	Components not packed by the applicable procedure for Level A or Level B.

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113. Strapping not as specified for Level A.
114. Quantities packed together exceed the size or weight limitation of the box for Level A or Level B.
115. Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. The BMS components shall be preserved by one of the following procedures depending upon procurement of complete BMS's or procurement of components separately.

5.1.1.1 Preservation of complete BMS's. The Magnetic Switch Assembly and Actuating Magnet Assembly that comprise a complete BMS shall each be preserved in accordance with MIL-P-116, Method IIc. The two components shall then be placed together in a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional or PPP-B-676, Variety 1, style, type, and class optional. The components shall be cushioned within the box to prevent movement or damage.

5.1.1.2 Major components furnished separately. Components procured separately shall be preserved and each placed in an individual box as specified in 5.1.1.1.

5.1.1.3 Consolidation. Components of like description, preserved as specified in 5.1.1.1 and 5.1.1.2 shall be placed together in a close-fitting box conforming to PPP-B-636, W5c, style optional, in quantities not to exceed the weight or size limitations of the box. Only boxes containing like contents shall be consolidated.

5.1.2 Level B. The BMS components shall be preserved as specified in 5.1.1 for Level A, except the method of preservation shall be Method IA-8.

5.1.3 Commercial. Each complete BMS and components procured separately shall be preserved in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The BMS components preserved as specified in 5.1 shall be packed by one of the following procedures depending upon procurement of complete BMS's or components procured separately.

5.2.1.1 Packing of complete BMS's. Complete BMS's shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional in quantities not to exceed the weight limitations of the box. Closure and strapping shall be in accordance with the appendix of the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.1.2 Packing of major components when procured separately. BMS components of like description shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete BMS's or BMS components (when procured separately) and preserved as specified in 5.1 shall be packed as specified in 5.2.1.1 or 5.2.1.2 as applicable, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Complete BMS and BMS components (when procured separately) and preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for Level A or Level B shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The label shall contain the Project Code "JID".

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6. NOTES

6.1 Intended use. The BMS is a sensing device intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When major components are to be furnished separately, and number to be furnished (see 3.1.1).
(c) Time frame required for submission of preproduction model and number of BMS's and major components (when procured separately) to be furnished (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examinations and tests. When the Government will conduct some but not all of the preproduction examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of BMS's to be furnished, when applicable (see 3.3).
(f) When cleaning, treating, painting, and color shall be other than as specified (see 3.8).
(g) When Government-loaned property is not required (see 3.9).
(h) Degree of preservation and packaging and level of packing required (see 5.1 and 5.2).
(i) When other than Finish B strapping is required (see 5.2.1.1).

6.3 Preproduction model. Any changes or deviations of production BMS's and major components (when procured separately) from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish BMS's and major components (when procured separately) conforming to this specification.

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6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the BMS is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component; physical change which impairs the mechanical integrity of the BMS; evidence of delamination or water penetration into integrated circuits; printed circuit boards; or any physical condition resulting in noncompliance of the units with the provisions of this specification.

6.4.3 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.9, when applicable (see 6.2).

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each BMS.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of BMS's, or major components procured separately.

6.8 Supplementary quality assurance provisions (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

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6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.4).

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
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APPENDIX

TEST METHODS

FOR

BALANCED MAGNETIC SWITCH

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether BMS's and major components procured under this specification conform to the requirements specified.

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TEST METHOD 1

FUNCTIONAL TEST

1. TESTS TO BE PERFORMED:

(a) Alarm/no alarm (field strength changes) requirement.
(b) Tamper protection requirement.

2. TEST APPARATUS:

(a) Multimeter - Simpson 260 or equivalent.
(b) Test fixture to supply a uniform magnetic field (Helmholtz Coil method).

3. TEST PROCEDURE:

(a) Alarm/no alarm (field strength change). The BMS shall satisfy the following requirements when the adjustments are made as indicated:

(1) Position the two major components such that their longitudinal axes are parallel and in a common horizontal plane. Connect an ohmmeter between Terminals 1 and 2 of TB1.

(2) Position Actuating Magnet Assembly 0.250 inch away from active face of Magnetic Switch Assembly. The ohmmeter shall indicate 33 $\pm$ 4 ohms. If this condition does not exist, adjust the biasing magnet until an ohmmeter reading of 33 $\pm$ 4 ohm exists.

(3) Vertically move the Actuating Magnet Assembly parallel to the active face of the Magnetic Switch Assembly until a reading of more than 100K ohms is obtained on the ohmmeter. The vertical displacement of the actuating magnet must be more than 0.50 inch and less than 1.250 inches.

(4) Place the BMS in the Helmholtz coil assembly. Connect the coil to a power supply. Increase the current through the coil until the alarm condition is indicated by a reading of more than 100K ohms.

(5) Reverse the coil polarity and repeat Step 4.

(6) Determine the gauss level required to activate the switch assembly in each polarity deviation. Add the absolute values of the two gauss levels. The total shall be less the 67 gauss.

(7) Position Actuating Magnet Assembly 1-inch away from the active face of Magnetic Switch Assembly. Adjust the biasing magnet, if necessary, to obtain an ohmmeter reading of 33 $\pm$ 4 ohms.

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(8) Move the Actuating Magnet Assembly parallel to the active face of the Magnetic Switch Assembly until a reading of more than 100K ohms is obtained. The vertical displacement of the actuating magnet must be more than 0.50 inch and less than 1.250 inches.

(9) Repeat Steps 4 through 6 with the actuating magnet positioned as in Step 7.

(b) Tamper protection requirement.

(1) Connect the ohmmeter to TB1-5 and TB1-6, bringing the test leads through the conduit hole.

(2) Install cover and observe ohmmeter reading of <1 ohm.

(3) Lift the Magnetic Switch Assembly cover. An ohmmeter reading of more than 100K ohms shall occur with cover movement between 0.087 inch and 0.262 inch. Observe and verify that the cover flanges have not cleared the switch housing at this point.

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TEST METHOD 2

ACTUATING MAGNET ASSEMBLY

(13220E3301)

1. TEST APPARATUS:

(a) Gaussmeter - F. W. Bell; Model 615 or equivalent.
(b) Test fixture - Figure A-1.

2. TEST PROCEDURE:

(a) Perform the following gauss measurements. Measurement to be taken at the end of the magnet (see Figure A-1).

Measurement	Reading
Housing (C13220E3311-1)	1000-1700 gauss
Housing (C13220E3311-2)	1500-2000 gauss

(b) Verify north and south poles of both housings (see Figure A-1). North shall be at notched end of housing. A gaussmeter will read "+" for north pole with the meter switch in "NOR" position.

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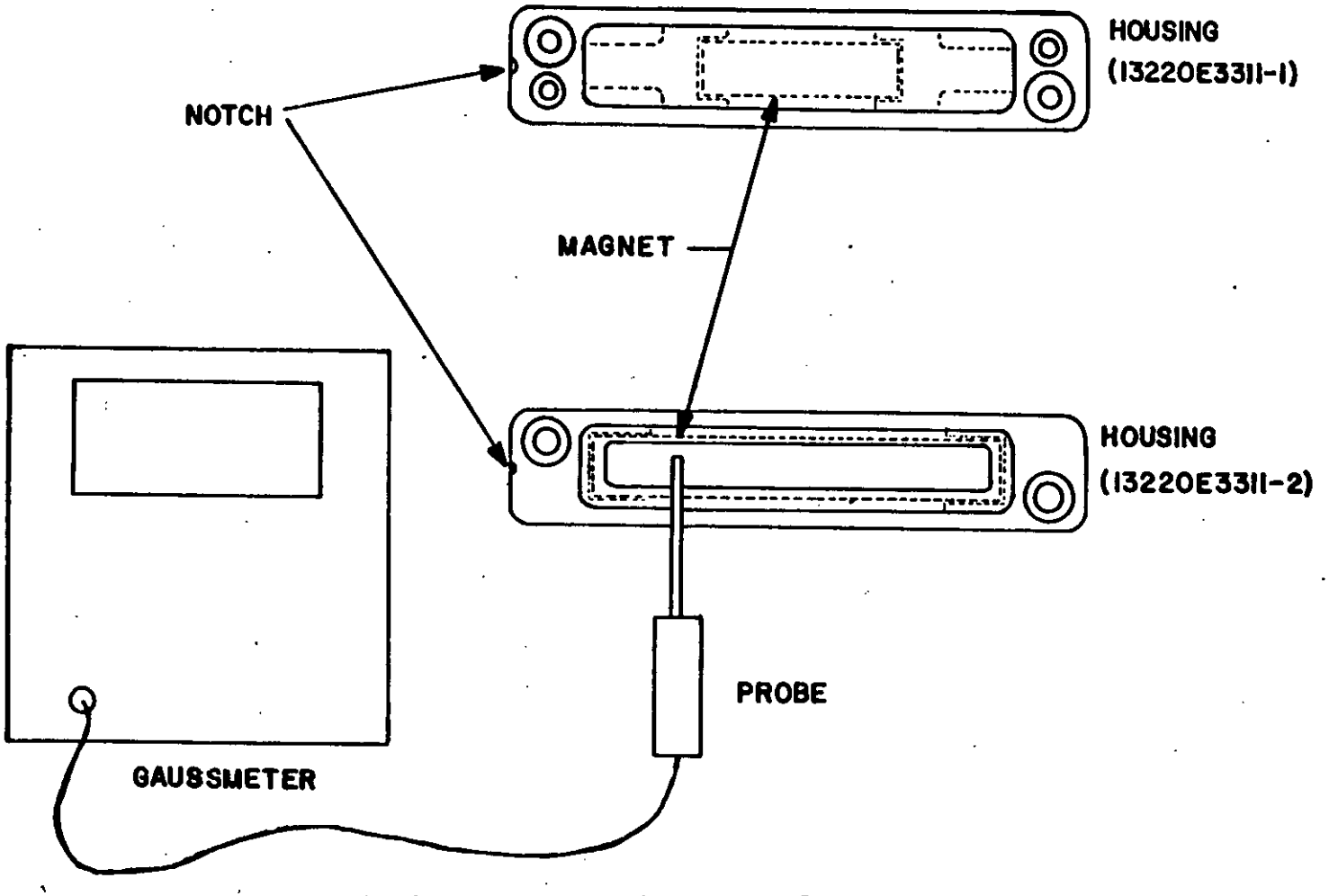


FIGURE A-1. Test fixture, actuating magnet assembly.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-52867B Switch, Balanced Magnetic, SA-1955/FSS-9(V)

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